Quantifying individual-level mosaic chromosomal alteration fitness: genetic determinants and clinical consequences

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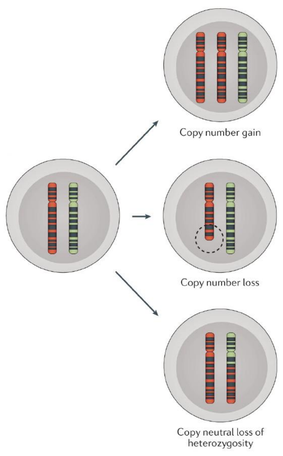
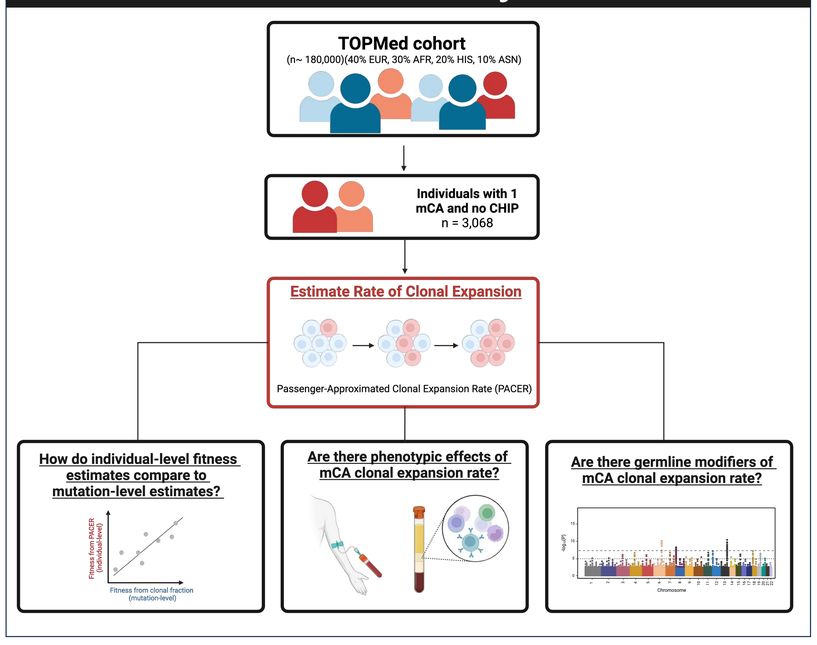
Clonal Hematopoiesis

- A clonal expansion of blood cells often caused by a genetic driver mutation in a hematopoietic stem cell.
- Linked to hematologic malignancy, cardiovascular disease, and infection susceptibility.
- May be caused by (1) **mosaic chromosomal alterations (e.g., mCAs)** or (2) point mutations (e.g., CHIP).
- **Clonal size matters**: large clones increase disease risk
- Determining clonal size with longitudinal samples limits sample size to determine drivers of mCA clonal expansion

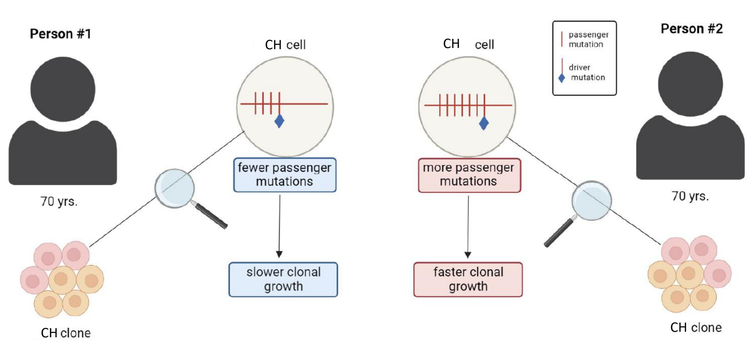


Our Study



The Concept of PACER[1]

- ***P**assenger-**A**pproximated **C**lonal **E**xpansion **R**ate* estimates clonal expansion rate **per-individual** from age, passenger mutation count, and clonal fraction at x38 coverage depth whole genome sequencing (WGS) of a single blood draw



- Given two individuals of the same age and with clones of the same size, the clone with more passengers will be more fit (i.e., expanded to the same size in less time).

We quantified clonal expansion rate for ~6k individuals with mCAs in the NHLBI TOPMed cohort from a single blood draw, identified associations between clonal growth rate and blood counts, and found germline genetic modulators of mCA fitness.

Variability in Clonal Fraction by mCA

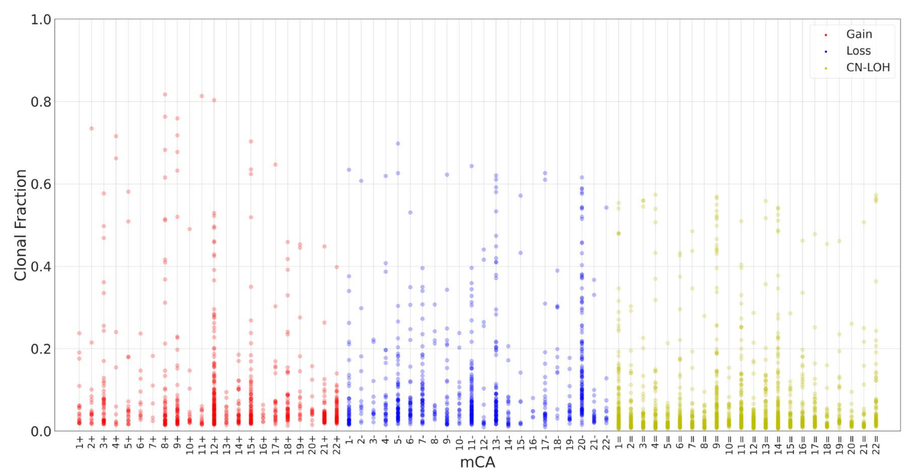


Fig 1. Autosomal mCA events observed in NHLBI TOPMed genomes. The clonal fractions for each patient with a specific mCA.

Validation of PACER Estimates

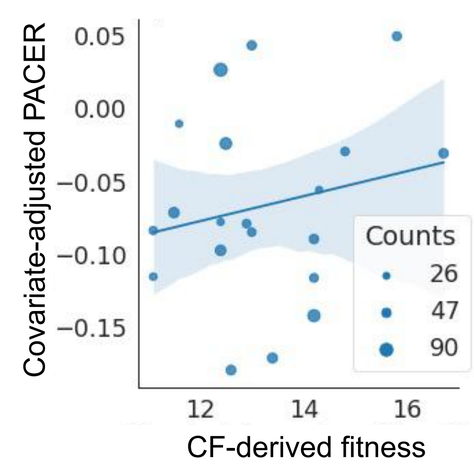


Fig 2. PACER vs clonal-fraction-derived fitness metric. Scatter plot of median total passenger mutations and fitness derived from clonal fraction by mCA.[2] **R^2 = 0.49.**

Comparison of mCA Fitness by Type

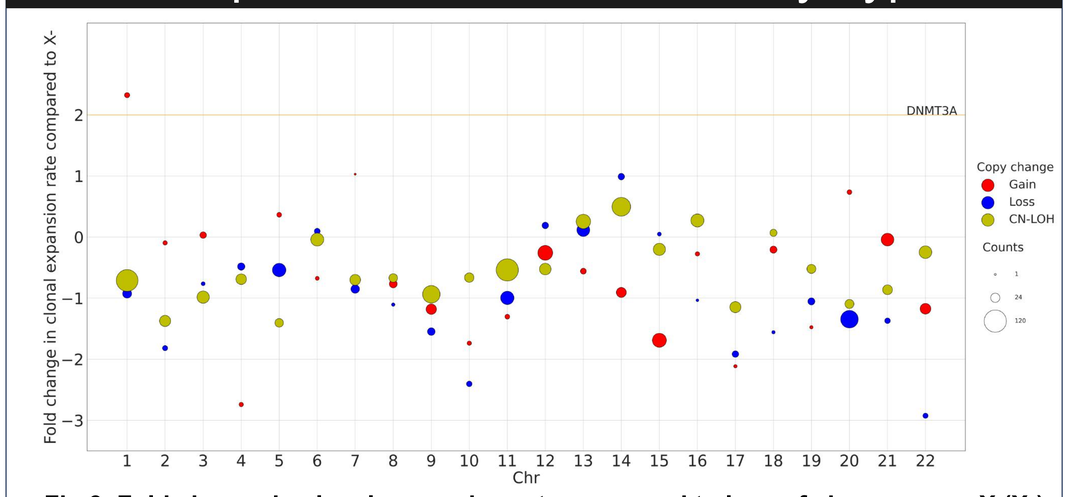
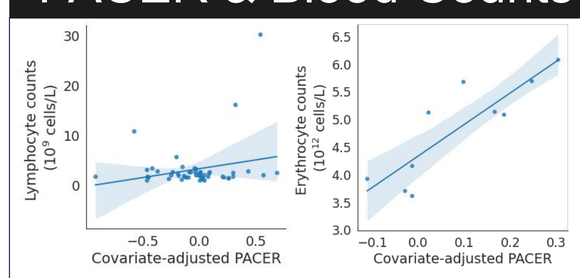


Fig 3. Fold change in clonal expansion rate compared to loss of chromosome X (X-) and DNMT3A CHIP. The clonal expansion rate is calculated after covariate adjustment (age, sex, clonal fraction) and inverse normalization of the total passenger mutations.

PACER & Blood Counts



mCA type	Cell counts	Beta [95% CI]	p
Lymphoid	lymphocytes	0.003 - 0.032	0.01*
Myeloid	myelocytes	-0.009 - 0.003	0.30
Erythroid	erythrocytes	0.006 - 0.018	0.02*

Fig 4. Blood counts and covariate-adjusted PACER. PACER was correlated with lymphocyte count among lymphoid mCAs (left) and erythrocyte count among chr9q mCAs (right) but not myeloid count among myeloid mCAs (not shown)[3].

Germline Associations with PACER and mCA Clone Prevalence

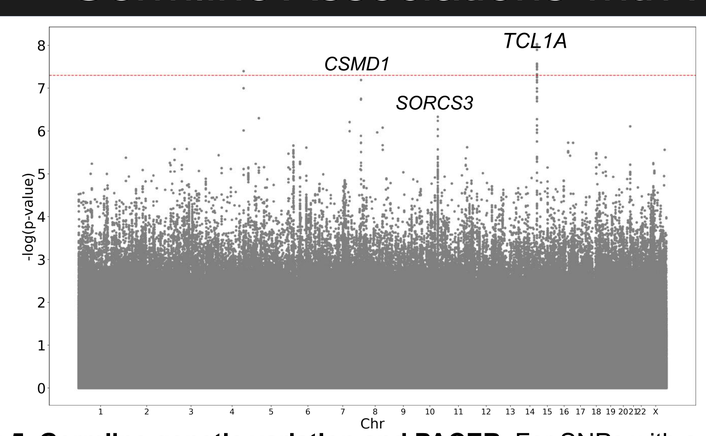


Fig 5. Germline genetic variation and PACER. For SNPs with a minor allele frequency > 1%. Regression of total passenger mutations with covariates. The dashed red line represents a p-value cutoff of 5×10^{-8}.

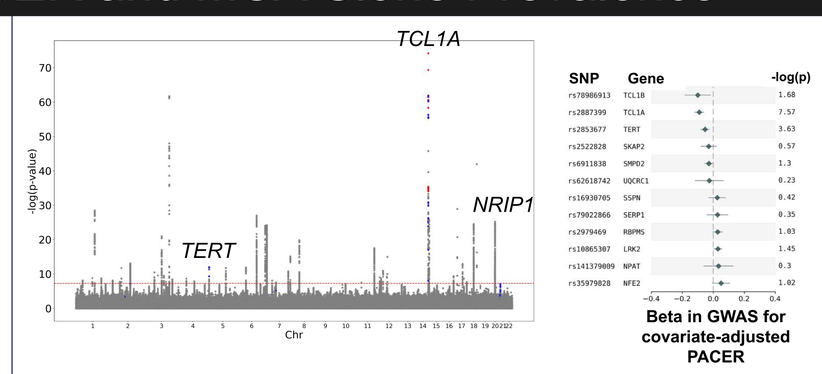


Fig 6. PACER GWAS elucidates mechanism of mCA prevalence GWAS. mCA clone prevalence GWAS[4] SNPs with a PACER GWAS p-value < 10^{-6} and 10^{-6} < p-value < 10^{-3}. Three loci act through modulating clonal expansion: TCL1A (chr14), TERT (chr 5), and NRIP1 (chr21).

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Conclusions

- PACER can reproducibly quantify mCA clonal expansion rate per individual from a single blood draw.
- mCAs have a lower clonal expansion rate than single nucleotide variant CHIP (e.g., DNMT3A).
- Lymphoid-cancer-associated mCAs had higher fitness than non-lymphoid mCAs.
- Among lymphoid-cancer-associated mCAs, PACER correlated with lymphocyte count.
- Variants in *TCL1A*, *TERT* and *NRIP1* modulate mCA clonal expansion rate and clone prevalence.

References

1. Weinstock JS et al. Aberrant activation of TCL1A promotes stem cell expansion in clonal haematopoiesis. Nature 616, 755–763 (2023).
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Acknowledgements

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